

# EVEN INTEGERS AS SUMS OF TWO SUMS OF TWO SQUARES WITH FEW PRIME FACTORS

RUQING CHEN

**ABSTRACT.** Let  $N \equiv 2 \pmod{4}$  be sufficiently large. We show that the number of  $n \leq N$  such that  $n$  and  $N - n$  are both sums of two squares and  $\Omega(n(N - n)) \leq 11$  is at least  $\frac{3}{2} \mathfrak{S}(N)N/(\log N)^2$ , where  $\mathfrak{S}(N)$  is the binary Goldbach singular series. The level of distribution of the sequence  $n(N - n)$  is 1, but sifting the primes  $q \equiv 3 \pmod{4}$  up to  $N^{1/2}$  with the linear sieve falls exactly on the sieving limit  $s = 2$ . We therefore sift these primes only up to  $N^{1/2-\delta}$  and remove the survivors with two large prime factors  $\equiv 3 \pmod{4}$  on the same side by a switching argument at the Bombieri–Vinogradov level. The final numerical inequality is verified with interval arithmetic. The same method gives  $\max(\Omega(n), \Omega(N - n)) \leq 8$  for a positive proportion of the expected number of  $n$ , and extends to the principal forms of all nine imaginary quadratic fields of class number one, with the bound 11 whenever  $N$  is not divisible by a high power of the ramified prime.

## 1. INTRODUCTION

Every positive integer is a sum of four squares, so every  $N$  is of the form  $n_1 + n_2$  with  $n_1, n_2$  sums of two squares. We ask for such representations in which  $n_1$  and  $n_2$  have few prime factors.

The multiplicative analogue, pairs  $n, n + 1$  that are both sums of two squares, was settled up to constants by Hooley [5] and Indlekofer [6], who showed that there are  $\asymp x/\log x$  such  $n \leq x$ ; correlations  $\sum r(n)r(n + h)$  of the representation function have since been studied by spectral methods, with uniformity in the coefficients of the forms, e.g. by Blomer [1]. On the side of almost prime variables, Brüdern and Fouvry [3] proved Lagrange’s four squares theorem with almost prime variables, and Blomer, Grimmelt, Li and Rydin Myerson [2] showed that every large  $n \equiv 1, 3 \pmod{6}$  is a prime plus two squares of almost primes. In these works the almost prime condition is on the variables  $x_i$ . Here it is on the values  $x^2 + y^2$  themselves. Finally, [4] treats the case where one summand is a prime and the other is an almost prime sum of two squares; the present problem is its two-sided analogue.

**Theorem 1.1.** *For every sufficiently large  $N \equiv 2 \pmod{4}$ ,*

$$\begin{aligned} \#\{n \leq N : n \equiv 1 \pmod{4}, n = x^2 + y^2, N - n = z^2 + w^2, \Omega(n(N - n)) \leq 11\} \\ \geq \frac{3}{2} \mathfrak{S}(N) \frac{N}{(\log N)^2}, \end{aligned}$$

where  $\mathfrak{S}(N) = 2 \prod_{\ell > 2} (1 - (\ell - 1)^{-2}) \prod_{\ell | N, \ell > 2} \frac{\ell - 1}{\ell - 2}$ .

*Remark 1.2* (The prime 2). The restriction  $N \equiv 2 \pmod{4}$  is natural. The odd part of a sum of two squares is  $\equiv 1 \pmod{4}$ . If  $N \equiv 0 \pmod{4}$ , both summands must contain powers of 2, and for  $N = 2^j$  the only representation is  $n_1 = n_2 = 2^{j-1}$ , so  $\Omega(n_1 n_2)$  is unbounded. For general  $N$  one should count prime factors of the odd parts only; we do not pursue this here.

*Remark 1.3.* With the parameters of the second line of Table 1 the same method gives the weaker bound  $\Omega(n(N - n)) \leq 12$  with the larger constant 2.24 in place of  $\frac{3}{2}$ ; both are certified.

The weights of Section 3 also control each side separately.

**Corollary 1.4.** *For every sufficiently large  $N \equiv 2 \pmod{4}$ , the number of  $n \leq N$ ,  $n \equiv 1 \pmod{4}$ , such that  $n$  and  $N - n$  are sums of two squares,  $\max(\Omega(n), \Omega(N - n)) \leq 8$  and  $\Omega(n(N - n)) \leq 12$ , is at least  $\frac{3}{5} \mathfrak{S}(N)N/(\log N)^2$ .*

The method is not specific to  $x^2 + y^2$ . Let  $\Delta < 0$  be a fundamental discriminant with class number  $h(\Delta) = 1$ , that is  $\Delta \in \{-3, -4, -7, -8, -11, -19, -43, -67, -163\}$ , let  $Q_\Delta$  be the principal binary quadratic form of discriminant  $\Delta$ , and let  $r$  be the prime ramified in  $\mathbb{Q}(\sqrt{\Delta})$ .

**Theorem 1.5.** *Let  $N$  be a sufficiently large even integer. Then*

$$\#\{n \leq N : n = Q_\Delta(x, y), N - n = Q_\Delta(z, w), \Omega(n(N - n)) \leq k_\Delta(N)\} \gg_\Delta \mathfrak{S}(N) \frac{N}{(\log N)^2},$$

where  $k_\Delta(N) = 11$  in each of the following cases:  $\Delta = -4$  and  $N \equiv 2 \pmod{4}$ ;  $\Delta = -8$  and  $v_2(N) \leq 2$ ;  $\Delta = -3$  and  $N \equiv 2 \pmod{3}$ ;  $\Delta = -r$  with  $r \geq 7$  and  $r \nmid N$ . Moreover  $k_{-3}(N) = 12$  if  $N \equiv 1 \pmod{3}$ , and  $k_\Delta(N) = 13$  in the cases  $\Delta = -4$ ,  $N \equiv 4 \pmod{8}$ ;  $\Delta = -8$ ,  $v_2(N) = 3$ ;  $\Delta = -3$ ,  $N \equiv 6 \pmod{9}$ ;  $\Delta = -r$  with  $r \geq 7$  and  $r \parallel N$ .

The extra prime factors in the second list are forced: they are powers of the ramified prime that every representation must contain (Lemma 5.1). The proof is given in Section 5.

**Method.** Restrict to  $n \equiv 1 \pmod{4}$ ; then  $N - n \equiv 1 \pmod{4}$ . If neither  $n$  nor  $N - n$  has a prime factor  $q \equiv 3 \pmod{4}$ , both are sums of two squares. The sequence  $\{n(N - n)\}$  has level of distribution  $N^{1-\varepsilon}$  (Lemma 2.1), and the primes  $q \equiv 3 \pmod{4}$  have two roots of  $n(N - n) \equiv 0 \pmod{q}$ , so they form a sifting set of dimension 1. Sifting them up to  $N^{1/2}$  at level  $N$  gives  $s = 2$ , where the linear lower bound function vanishes. We sift them up to  $y = N^{1/2-\delta}$  instead. Since  $n \equiv N - n \equiv 1 \pmod{4}$ , each side has an even number of such prime factors, so a survivor that is not a pair of sums of two squares has two large prime factors  $\equiv 3 \pmod{4}$  on the *same* side; these are counted from above by switching (Section 3). Prime factors are controlled by Richert's weights applied to  $n(N - n)$  jointly.

*Remark 1.6.* To the best of the author's knowledge, Theorems 1.1 and 1.5 have not appeared in the literature.

## 2. THE SEQUENCE $n(N - n)$

Let  $N \equiv 2 \pmod{4}$  be large,  $\mathcal{P}_1 = \{q \equiv 1 \pmod{4}\}$ ,  $\mathcal{P}_3 = \{q \equiv 3 \pmod{4}\}$ , and

$$\mathcal{A} = \{a_n = n(N - n) : 1 \leq n \leq N, n \equiv 1 \pmod{4}\}, \quad X = N/4.$$

For odd squarefree  $d$  let  $\rho(d)$  be the number of solutions of  $n(N - n) \equiv 0 \pmod{d}$ ; it is multiplicative with  $\rho(q) = 2$  for  $q \nmid N$  and  $\rho(q) = 1$  for  $q \mid N$ . Put  $g(d) = \rho(d)/d$ , so that

$$g(q) = \frac{2}{q} \quad (q \nmid N), \quad g(q) = \frac{1}{q} \quad (q \mid N),$$

and define  $r_d$  by  $\#\{n : a_n \in \mathcal{A}, d \mid a_n\} = Xg(d) + r_d$ .

**Lemma 2.1** (Level of distribution).  *$|r_d| \leq \rho(d)$ , and for  $D \leq N$ ,  $\sum_{d \leq D} \mu^2(d) \tau_3(d) |r_d| \ll D(\log D)^C$ . In particular the level of distribution is  $N^{1-\varepsilon}$ .*

*Proof.* The condition  $d \mid a_n$  together with  $n \equiv 1 \pmod{4}$  restricts  $n$  to  $\rho(d)$  classes modulo  $4d$ , each containing  $N/(4d) + O(1)$  integers  $n \leq N$ . Hence  $|r_d| \leq \rho(d) \leq 2^{\omega(d)}$ , and the sum is  $\ll \sum_{d \leq D} \tau_3(d) 2^{\omega(d)} \ll D(\log D)^C$ .  $\square$

**Lemma 2.2** (Local factors). *As  $z \rightarrow \infty$ ,*

$$\prod_{2 < q < z} (1 - g(q)) \sim \frac{\mathfrak{S}(N)}{2} \left( \frac{2e^{-\gamma}}{\log z} \right)^2.$$

*Proof.* For  $q \nmid N$ ,  $(1 - 2/q)(1 - 1/q)^{-2} = 1 - (q - 1)^{-2}$ ; for  $q \mid N$ ,  $(1 - 1/q)(1 - 1/q)^{-2} = q/(q - 1)$ . The product of these factors over odd  $q$  converges to  $\mathfrak{S}(N)/2$ , and  $\prod_{2 < q < z} (1 - 1/q) \sim 2e^{-\gamma}/\log z$ .  $\square$

We use the following sieve tools; see [4] for statements and references. The linear sieve ( $\kappa = 1$ ,  $\beta = 2$ ) has functions  $F_1, f_1$  with  $F_1(s) = 2e^\gamma/s$  for  $0 < s \leq 3$  and  $f_1(s) = 2e^\gamma \log(s - 1)/s$  for  $2 \leq s \leq 4$ ,  $f_1(s) = 0$  for  $s \leq 2$  [8]; the semi-linear sieve ( $\kappa = 1/2$ ,  $\beta = 1$ ) has functions  $F_{1/2}, f_{1/2}$  [7]. The vector sieve inequality [3] is  $ab \geq a^-b^+ + a^+b^- - a^+b^+$  for  $a, b \in \{0, 1\}$ ,  $a^- \leq a \leq a^+$ ,  $b^- \leq b \leq b^+$ ,  $a^+, b^+ \geq 0$ . The Bombieri–Vinogradov theorem for convolutions is used as in [4, Lemma 2.4].

Figure 1 shows  $F_1$  and  $f_1$ . Sifting the primes  $q \equiv 3 \pmod{4}$  up to  $N^{1/2}$  at level  $N$  would give  $s = 2$ , where  $f_1$  vanishes; with  $y = N^{1/2-\delta}$  the working point moves to  $s_I = \theta_I/(1/2 - \delta)$ , which is  $\approx 3.62$  for the parameters of Theorem 1.1.

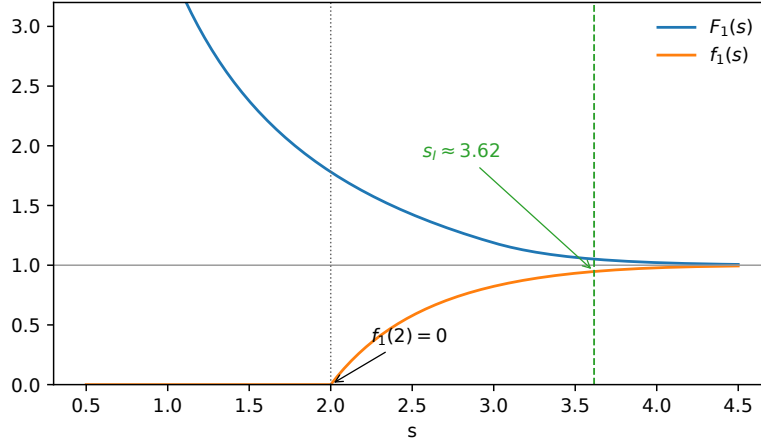


FIGURE 1. The linear sieve functions  $F_1(s)$  and  $f_1(s)$  ( $\kappa = 1$ ,  $\beta = 2$ ). The lower bound function vanishes for  $s \leq 2$ . The dashed line marks the working point  $s_I = 0.9220/(1/2 - 0.245) \approx 3.62$  of the inert sieve in the proof of Theorem 1.1.

### 3. THE WEIGHTED SIEVE AND SWITCHING

Fix  $0 < \alpha < \delta < 1/4$ ,  $0 < \lambda$ ,  $\alpha < \beta_1 \leq 1/2$ , and put

$$y = N^{1/2-\delta}, \quad z = N^\alpha, \quad y_1 = N^{\beta_1}, \quad P_I = \prod_{\substack{q \in \mathcal{P}_3 \\ q < y}} q, \quad P_S = \prod_{\substack{q \in \mathcal{P}_1 \\ q < z}} q, \quad P = P_I P_S.$$

Every odd prime  $q < z$  divides  $P$ . For  $a_n \in \mathcal{A}$  with  $(a_n, P) = 1$  define

$$w_n = 1 - \lambda \sum_{\substack{q|a_n, q \in \mathcal{P}_1 \\ z \leq q < y_1}} \left( 1 - \frac{\log q}{\log y_1} \right), \quad W = \sum_{(a_n, P)=1} w_n.$$

**Lemma 3.1** (Structure). *Let  $(a_n, P) = 1$ . Then either*

- (i)  $n$  and  $N - n$  are both sums of two squares, all their prime factors are  $\geq z$ , and those in  $\mathcal{P}_3$  occur only as squares  $q^2$ ; or
- (ii) for  $m' = n$  or  $m' = N - n$  we have  $m' = q_1 q_2 m$  with  $q_1 < q_2$ ,  $q_1, q_2 \in \mathcal{P}_3$ ,  $q_1 \geq y$ , and  $m \leq N^{2\delta}$  composed of primes of  $\mathcal{P}_1$  that are  $\geq z$ .

*Proof.* Each of  $n$ ,  $N - n$  is odd and  $\equiv 1 \pmod{4}$ , so each has an even number of prime factors in  $\mathcal{P}_3$ , counted with multiplicity. They are all  $\geq y$ , and  $y^4 > N$ , so each side has 0 or 2 of them. If on both sides there are none, or they coincide, both sides are sums of two squares. Otherwise some side is as in (ii). In particular, two exceptional prime factors never lie on different sides.  $\square$

**Lemma 3.2** (Prime factors). *If  $a_n$  is as in Lemma 3.1(i), squarefree, and  $w_n > 0$ , then  $\Omega(n(N - n)) < 1/\lambda + 2/\beta_1$  and  $\max(\Omega(n), \Omega(N - n)) < 1/\lambda + 1/\beta_1$ .*

*Proof.* All prime factors of  $a_n$  are in  $\mathcal{P}_1$  and  $\geq z$ , and since  $a_n$  is squarefree each of them divides exactly one of  $n$ ,  $N - n$ . Write  $w_n = 1 - \lambda(T_1 + T_2)$ , where  $T_1, T_2$  are the sums over  $q \mid n$  and  $q \mid N - n$ . Each  $T_i$  is a sum of non-negative terms, and as in [4, Lemma 3.2],  $T_1 \geq \Omega(n) - \log n / \log y_1 \geq \Omega(n) - 1/\beta_1$ , and similarly for  $T_2$ . Now  $w_n > 0$  gives  $T_1 + T_2 < 1/\lambda$ , which implies both  $\Omega(n(N - n)) - 2/\beta_1 < 1/\lambda$  and, since  $T_2 \geq 0$ ,  $\Omega(n) - 1/\beta_1 < 1/\lambda$ ; the same holds for  $N - n$ .  $\square$

Non-squarefree  $a_n$  with a square factor  $q^2$ ,  $q \geq z$ , number  $\ll N^{1-\alpha}$  and are negligible.

**Proposition 3.3** (Main term). *Let  $\theta_I \in (0, 1)$  and, for  $u \in [\alpha, \beta_1]$ ,  $t(u) \in (0, 1 - u)$ . Put  $s_I = \theta_I/(1/2 - \delta)$ ,  $s_S = (1 - \theta_I)/\alpha$ ,*

$$L = f_1(s_I)F_1(s_S) + F_1(s_I)f_1(s_S) - F_1(s_I)F_1(s_S),$$

$$R = \int_{\alpha}^{\beta_1} \left(1 - \frac{u}{\beta_1}\right) F_1\left(\frac{t(u)}{1/2 - \delta}\right) F_1\left(\frac{1 - u - t(u)}{\alpha}\right) \frac{du}{u}.$$

*Then  $W \geq (L - \lambda R + o(1)) XV$ , where  $V = \prod_{q \mid P} (1 - g(q))$  and*

$$XV \sim \frac{e^{-2\gamma}}{2\alpha(1/2 - \delta)} \mathfrak{S}(N) \frac{N}{(\log N)^2}.$$

*Proof.* Apply the vector sieve with  $a = 1_{(a_n, P_I)=1}$  and  $b = 1_{(a_n, P_S)=1}$ . Both are linear sieves:  $P_I$  and  $P_S$  consist of primes from one half of the residue classes modulo 4, each with  $\rho(q) = 2$  (apart from the finitely many  $q \mid N$  below a given bound, which do not affect the dimension). Use levels  $N^{\theta_I}$  and  $N^{1-\theta_I-\varepsilon}$ ; the combined moduli are  $\leq N^{1-\varepsilon}$  and Lemma 2.1 controls the remainders. For the weights,  $q \mid a_n$  for  $q \in \mathcal{P}_1$ ,  $q \nmid N$ , has density  $g(q) = 2/q$ , and  $\sum_{q \in \mathcal{P}_1} 2/q$  over  $N^u \leq q < N^{u+du}$  is  $du/u$ . The upper bound vector sieve for  $\mathcal{A}_q$  at levels  $N^{t(u)}$ ,  $N^{1-u-t(u)-\varepsilon}$  gives  $R$ . Finally  $V = \prod_{2 < q < z} (1 - g(q)) \prod_{q \in \mathcal{P}_3, z \leq q < y} (1 - g(q))$ , and Lemma 2.2 together with  $\prod_{q \in \mathcal{P}_3, z \leq q < y} (1 - 2/q) \sim \log z / \log y$  gives the asymptotic for  $XV$ .  $\square$

**Proposition 3.4** (Switching). *Let  $\mathcal{E}$  be the set of  $n$  with  $(a_n, P) = 1$  in case (ii) of Lemma 3.1. Then*

$$\#\mathcal{E} \leq \left(4c(\delta, \alpha) e^{\gamma} \sqrt{\alpha(1/2 - \delta)} U + o(1)\right) XV,$$

*where  $c(\delta, \alpha)$  is the switching density of [4, Prop. 4.1] and*

$$U = \min_{0 < t < 1/2} F_{1/2}\left(\frac{t}{1/2 - \delta}\right) F_{1/2}\left(\frac{1/2 - t}{\alpha}\right).$$

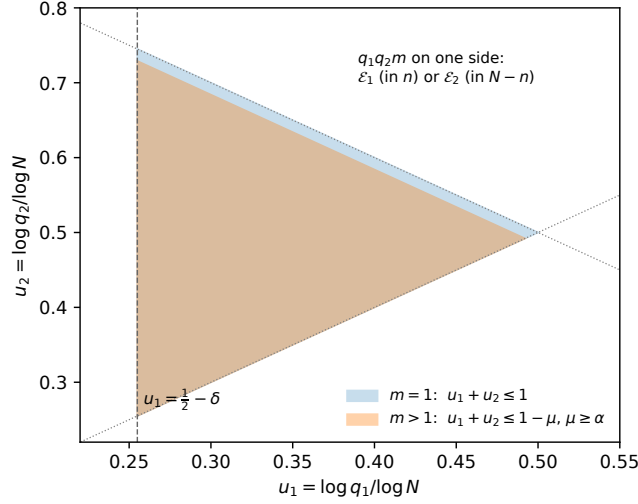


FIGURE 2. The switching region for  $\delta = 0.245$ ,  $\alpha = 0.016$ , in the coordinates  $u_i = \log q_i / \log N$ :  $\frac{1}{2} - \delta \leq u_1 < u_2$  and  $u_1 + u_2 \leq 1 - \mu$ , where  $m = N^\mu$  with  $\mu = 0$  ( $m = 1$ ) or  $\mu \geq \alpha$ . By Lemma 3.1 the triple  $q_1 q_2 m$  lies entirely in  $n$  (the set  $\mathcal{E}_1$ ) or entirely in  $N - n$  (the set  $\mathcal{E}_2$ ).

*Proof.* Write  $\mathcal{E} = \mathcal{E}_1 \cup \mathcal{E}_2$  according as the exceptional pair lies in  $n$  or in  $N - n$ ; the two cases are symmetric, which accounts for a factor 2 (Figure 2). Consider  $\mathcal{E}_1$ . Let  $\mathcal{T}$  be the set of triples  $(q_1, q_2, m)$  as in Lemma 3.1(ii) with  $t = q_1 q_2 m \leq N$ , so that  $|\mathcal{T}| = (c(\delta, \alpha) + o(1))N / \log N$  as in [4]; note  $t \equiv 1 \pmod{4}$  automatically. Every  $n \in \mathcal{E}_1$  equals such a  $t$ , and  $N - t$  has no prime factor in  $P$ . Hence  $\#\mathcal{E}_1$  is at most the number of elements of  $\mathcal{B} = \{N - t : t \in \mathcal{T}\}$  coprime to  $P$ .

Let  $\ell \mid P$ . No prime factor of  $t$  divides  $P$  (the  $q_i$  are  $\geq y$  and the prime factors of  $m$  are split and  $\geq z$ ), so  $t$  is equidistributed in the nonzero classes modulo  $\ell$ . Hence, for  $\ell \nmid N$ ,  $\ell \mid N - t$  with density  $g_{\mathcal{B}}(\ell) = 1/(\ell - 1)$ , while for  $\ell \mid N$  we have  $\ell \nmid N - t$ , so  $g_{\mathcal{B}}(\ell) = 0$ . The upper bound vector sieve for  $\mathcal{B}$  with the semi-linear components  $P_I$  and  $P_S$  at total level  $N^{1/2}(\log N)^{-B}$  (Bombieri–Vinogradov for the convolution  $q_1 * (q_2 m)$ ) gives

$$\#\mathcal{E}_1 \leq (U + o(1)) |\mathcal{B}| \prod_{q \mid P} (1 - g_{\mathcal{B}}(q)).$$

Now  $(1 - 1/(\ell - 1))(1 - 1/\ell)^{-1} = 1 - (\ell - 1)^{-2}$  and  $(1 - 0)(1 - 1/\ell)^{-1} = \ell/(\ell - 1)$  for  $\ell \mid N$ : these are the same local factors as in Lemma 2.2. Hence

$$\prod_{q \mid P} (1 - g_{\mathcal{B}}(q)) \sim \frac{\mathfrak{S}(N)}{2} \cdot \frac{2e^{-\gamma}}{\log z} \left( \frac{\log z}{\log y} \right)^{1/2},$$

and therefore

$$\#\mathcal{E}_1 \leq (U + o(1)) c \frac{N}{\log N} \cdot \frac{\mathfrak{S}(N)e^{-\gamma}}{\alpha \log N} \left( \frac{\alpha}{1/2 - \delta} \right)^{1/2} = (2ce^{\gamma} \sqrt{\alpha(1/2 - \delta)} U + o(1)) XV,$$

using Proposition 3.3 for  $XV$ . The singular series cancels exactly. Adding  $\mathcal{E}_2$  gives the claim.  $\square$

## 4. NUMERICAL VERIFICATION AND PROOF OF THEOREM 1.1

By Lemma 3.1 and  $w_n \leq 1$ , the weighted count over case (i) is at least  $W - \#\mathcal{E} \geq (\text{net} + o(1))XV$  with

$$\text{net} = L - \lambda R - 4c(\delta, \alpha)e^\gamma \sqrt{\alpha(1/2 - \delta)} U.$$

By Lemma 3.2, every squarefree counted  $n$  with  $w_n > 0$  satisfies  $\Omega(n(N - n)) < 1/\lambda + 2/\beta_1$  and  $\max(\Omega(n), \Omega(N - n)) < 1/\lambda + 1/\beta_1$ . We use the parameter sets of Table 1; the quantities  $1/\lambda + 2/\beta_1$  ( $= 12, 13, 3280/267$ ) and  $1/\lambda + 1/\beta_1$  ( $= 2390/267 < 9$  in the third line) are computed in exact rational arithmetic.

| $\delta$ | $\alpha$ | $\lambda$ | $\beta_1$ | $k$    | $L \geq$ | $R \leq$ | $c \leq$ | $U \leq$ | $\text{net} \geq$ |
|----------|----------|-----------|-----------|--------|----------|----------|----------|----------|-------------------|
| 0.245    | 0.016    | 1/8       | 1/2       | 11     | 0.944016 | 2.830154 | 1.067736 | 1.134620 | 0.038949          |
| 0.22     | 0.010    | 1/8       | 2/5       | 12     | 0.913373 | 3.144708 | 1.092388 | 1.166735 | 0.039811          |
| 0.24     | 0.022    | 89/500    | 3/10      | 12 (8) | 0.919835 | 1.939896 | 0.878693 | 1.166966 | 0.022031          |

TABLE 1. Certified bounds. Here  $k = \lceil 1/\lambda + 2/\beta_1 \rceil - 1$  bounds  $\Omega(n(N - n))$ ; in the third line (8) is the bound  $\lceil 1/\lambda + 1/\beta_1 \rceil - 1$  for  $\max(\Omega(n), \Omega(N - n))$ . The linear sieve level split is  $\theta_I = 0.9220, 0.9495, 0.8965$ .

**Lemma 4.1.** *The bounds of Table 1 hold.*

*Method of verification.* The computation is carried out by the script `code/certify.py`.

- (1) The sieve functions  $F_1, f_1$  ( $\kappa = 1$ ) and  $F_{1/2}, f_{1/2}$  ( $\kappa = 1/2$ ) are evaluated from their closed forms on  $(0, 3], [2, 4]$  resp.  $(0, 2], [1, 3]$  in interval arithmetic (`mpmath.iv`, 80 bits). Beyond these ranges the delay-differential equations  $(s^\kappa F(s))' = \kappa s^{\kappa-1} f(s-1)$ ,  $(s^\kappa f(s))' = \kappa s^{\kappa-1} F(s-1)$  are integrated on the grid  $10^{-3}\mathbb{Z}$  (up to  $s = 110$  resp. 60): on each step the lagged function is enclosed between its values at the end points, using that  $F$  decreases and  $f$  increases, and  $\int \kappa s^{\kappa-1} ds$  is computed exactly. Each floating point operation is followed by one outward rounding step. Off the grid, monotonicity gives one-sided bounds.
- (2)  $L$  is bounded below via  $L = f_1(s_I)F_1(s_S) + F_1(s_I)(f_1(s_S) - F_1(s_S))$  and  $f_1(s_S) - F_1(s_S) \leq 0$ .
- (3)  $R$  is bounded above by a Riemann sum over 4000 cells with  $t(u)$  constant on each cell; on a cell  $[u_a, u_b]$  the factor  $(1 - u/\beta_1)/u$  is largest at  $u_a$  and  $F_1((1 - u - t)/\alpha)$  is largest at  $u_b$ .  $U$  is bounded above by evaluating the product at one admissible  $t$ .
- (4) For  $c$ , the measure  $\frac{1}{2} 1_{[\alpha, 2\delta]}(\mu) d\mu/\mu$  is replaced by the masses  $\frac{1}{2} \log \frac{k+1}{k}$  placed at the left end points  $k/2000$ . The integrand  $I(\mu) = \frac{1}{1-\mu} \log \frac{1/2+\delta-\mu}{1/2-\delta}$  is decreasing on  $[0, 2\delta]$ : its derivative has the sign of  $\log \frac{a-\mu}{b} - \frac{1-\mu}{a-\mu}$  with  $a = \frac{1}{2} + \delta < 1$ ,  $b = \frac{1}{2} - \delta$ , whose first term decreases and second increases, and which is negative at  $\mu = 0$  (checked in the script). So moving mass to the left gives an upper bound. The floating point convolution is multiplied by  $1 + 10^{-9}$ .

A finer independent computation of  $c(0.22, 0.01)$  (grid  $1/20000$ ) gives 1.0913, consistent with the certified upper bound.  $\square$

*Proof of Theorem 1.1.* Take the first line of Table 1. By Lemma 4.1 and Proposition 3.3, after removing the  $\ll N^{1-\alpha}$  non-squarefree  $a_n$ , the number of counted  $n$  is at least

$$(0.038949 + o(1)) \frac{e^{-2\gamma}}{2 \cdot 0.016 \cdot 0.255} \mathfrak{S}(N) \frac{N}{(\log N)^2} \geq \frac{3}{2} \mathfrak{S}(N) \frac{N}{(\log N)^2}$$

| $\Delta$                 | $Q_\Delta$                         | $k_\Delta(N) = 11$    | $k_\Delta(N) = 12, 13$                                |
|--------------------------|------------------------------------|-----------------------|-------------------------------------------------------|
| -3                       | $x^2 + xy + y^2$                   | $N \equiv 2 \pmod{3}$ | 12: $N \equiv 1 \pmod{3}$ ; 13: $N \equiv 6 \pmod{9}$ |
| -4                       | $x^2 + y^2$                        | $N \equiv 2 \pmod{4}$ | 13: $N \equiv 4 \pmod{8}$                             |
| -7                       | $x^2 + xy + 2y^2$                  | $7 \nmid N$           | 13: $7 \parallel N$                                   |
| -8                       | $x^2 + 2y^2$                       | $v_2(N) \leq 2$       | 13: $v_2(N) = 3$                                      |
| -11, -19, -43, -67, -163 | $x^2 + xy + \frac{1-\Delta}{4}y^2$ | $r \nmid N$           | 13: $r \parallel N$                                   |

TABLE 2. The bound  $k_\Delta(N)$  of Theorem 1.5 ( $N$  even).

for large  $N$  (the numerical constant is  $\geq 1.5046$ ). Each of them has  $n$  and  $N - n$  sums of two squares and  $\Omega(n(N - n)) \leq 11$ .  $\square$

*Proof of Corollary 1.4.* The same argument with the third line of Table 1 gives at least

$$(0.022031 + o(1)) \frac{e^{-2\gamma}}{2 \cdot 0.022 \cdot 0.26} \mathfrak{S}(N) \frac{N}{(\log N)^2} \geq \frac{3}{5} \mathfrak{S}(N) \frac{N}{(\log N)^2}$$

counted  $n$  (the numerical constant is  $\geq 0.607$ ), and Lemma 3.2 gives  $\max(\Omega(n), \Omega(N - n)) \leq 8$  and  $\Omega(n(N - n)) \leq 12$ .  $\square$

## 5. OTHER CLASS NUMBER ONE FORMS

Let  $\Delta$ ,  $Q_\Delta$ ,  $r$  be as in Theorem 1.5 and  $\chi = \chi_\Delta$  the Kronecker symbol. An integer  $m \geq 1$  is represented by  $Q_\Delta$  if and only if every prime  $q$  with  $\chi(q) = -1$  divides  $m$  to an even power; the represented integers are the norms from  $\mathcal{O}_{\mathbb{Q}(\sqrt{\Delta})}$  and form a multiplicative set containing  $r$  [4, Lemma 6.1]. For  $u$  coprime to  $\Delta$ ,  $\chi(u) = (-1)^{\#\{\text{inert prime factors of } u\}}$  (with multiplicity).

**Lemma 5.1** (Admissible classes). *Let  $N$  be even and in one of the cases of Theorem 1.5, and let  $\kappa_\Delta(N) = k_\Delta(N) - 11$ . There are a modulus  $M$ , divisible by 2 and by  $r$ , and a class  $a \pmod{M}$  such that every  $n \equiv a \pmod{M}$  satisfies  $n = eu$ ,  $N - n = e'u'$ , where  $e, e'$  are fixed powers of  $r$  with  $\Omega(ee') = \kappa_\Delta(N)$ , the integers  $u, u'$  are coprime to  $2r$ , and  $\chi(u) = \chi(u') = 1$ . Moreover,  $r$  divides  $M/e$  and  $M/e'$ .*

*Proof.* Let  $r$  be odd, so  $\Delta = -r$  with  $r \equiv 3 \pmod{4}$  and  $\chi(u) = (u/r)$  for odd  $u$ . For  $N' \not\equiv 0 \pmod{r}$ ,

$$\#\{t \pmod{r} : (t/r) = ((N' - t)/r) = 1\} = \frac{1}{4} \sum_{t \neq 0, N'} \left(1 + \left(\frac{t}{r}\right)\right) \left(1 + \left(\frac{N' - t}{r}\right)\right) = \frac{r - 1 - 2(N'/r)}{4},$$

since  $\sum_t \left(\frac{t(N' - t)}{r}\right) = -\left(\frac{-1}{r}\right) = 1$ . This is  $\geq 1$  for  $r \geq 7$ , and for  $r = 3$  it is 1 if  $N' \equiv 2$  and 0 if  $N' \equiv 1 \pmod{3}$ . If  $r \nmid N$ , take  $N' = N$  and  $e = e' = 1$ . If  $r \parallel N$ , then  $(n/r) = ((N - n)/r) = 1$  with  $r \nmid n$  is impossible because  $(-1/r) = -1$ , so  $r$  divides both sides; take  $e = e' = r$  and  $N' = N/r$ . For  $\Delta = -3$  and  $N \equiv 1 \pmod{3}$  take  $e = 3$ ,  $e' = 1$ ,  $n \equiv 3 \pmod{9}$ ; then  $u \equiv 1$  and  $N - n \equiv 1 \pmod{3}$ . For  $\Delta = -3$ ,  $N \equiv 6 \pmod{9}$  take  $e = e' = 3$  and  $N' = N/3 \equiv 2$ . In all cases add the condition  $n$  odd. For  $\Delta = -4$ : if  $N \equiv 2 \pmod{4}$  take  $n \equiv 1 \pmod{4}$ ; if  $N \equiv 4 \pmod{8}$  take  $n \equiv 2 \pmod{8}$ , so that  $n/2 \equiv N/2 - n/2 \equiv 1 \pmod{4}$ . For  $\Delta = -8$ , where  $\chi(u) = 1$  iff  $u \equiv 1, 3 \pmod{8}$ : if  $v_2(N) \leq 2$  take  $n \equiv 1$  or  $3 \pmod{8}$  with  $N - n \equiv 1$  or  $3 \pmod{8}$ ; if  $v_2(N) = 3$  take  $n \equiv 2 \pmod{16}$ , so that  $n/2 \equiv 1$  and  $(N - n)/2 \equiv 3 \pmod{8}$ . We have checked by enumeration (script `code/local.conditions.py`) that these are the minimal costs.  $\square$

*Proof of Theorem 1.5.* Let  $M, a, e, e'$  be as in Lemma 5.1,  $\mathcal{A}_\Delta = \{a_n = n(N - n) : n \leq N, n \equiv a \pmod{M}\}$  and  $X = N/M$ . For  $d$  coprime to  $M$ ,  $d \mid a_n$  restricts  $n$  to  $\rho(d)$  classes modulo  $Md$ , so

Lemma 2.1 holds unchanged. We sift by the primes  $q \nmid M$ , with  $\mathcal{P}_3$ ,  $\mathcal{P}_1$  replaced by the inert and split primes; both have density  $1/2$  by Dirichlet's theorem modulo  $|\Delta|$ . Lemma 3.1 holds for  $u, u'$  (each has an even number of inert prime factors, all  $\geq y$ , and  $y^4 > N$ ), and Lemma 3.2 applied to  $uu'$  gives  $\Omega(uu') \leq 11$ , so  $\Omega(n(N-n)) \leq 11 + \kappa_\Delta(N)$ .

It remains to check the normalisation. Put  $\mathfrak{S}_M(N) = \prod_{q \nmid M} (1 - g(q))(1 - 1/q)^{-2}$ . Since  $\prod_{q < z, q \nmid M} (1 - 1/q) \sim \frac{M}{\varphi(M)} \frac{e^{-\gamma}}{\log z}$ ,

$$XV \sim \frac{M}{\varphi(M)^2} \mathfrak{S}_M(N) \frac{e^{-2\gamma}}{\alpha(1/2 - \delta)} \frac{N}{(\log N)^2}.$$

In the switching step for the side  $n = et$ , the triples  $t = q_1 q_2 m$  satisfy  $\chi(t) = 1$  automatically and must lie in one prescribed class modulo  $M/e$  with  $\chi = 1$ ; by the prime number theorem in progressions they are equidistributed among the  $\varphi(M/e)/2$  such classes, and  $t \leq N/e$ . Since  $r \mid M/e$  we have  $e\varphi(M/e) = \varphi(M)$ , so  $|\mathcal{T}| = (2/\varphi(M) + o(1))c(\delta, \alpha)N/\log N$ . For  $\ell \nmid M$ ,  $\ell \mid N - et$  with density  $1/(\ell - 1)$  if  $\ell \nmid N$  and 0 if  $\ell \mid N$ ; these are the local factors of  $\mathfrak{S}_M(N)$ , and

$$\prod_{q \mid P} (1 - g_{\mathcal{B}}(q)) \sim \frac{M}{\varphi(M)} \mathfrak{S}_M(N) \frac{e^{-\gamma}}{\log z} \left( \frac{\log z}{\log y} \right)^{1/2}.$$

Hence the contribution of this side is at most  $(2ce^\gamma \sqrt{\alpha(1/2 - \delta)}U + o(1))XV$ , exactly as in Proposition 3.4, and the same holds for the side  $N - n = e't$ . The Richert terms scale with  $XV$  as before. Therefore the normalised quantity is the same net as in Section 4, independently of  $\Delta$  and of the case, and Lemma 4.1 gives a count  $\geq (0.0389 + o(1))XV \gg_\Delta \mathfrak{S}(N)N/(\log N)^2$ , since  $\mathfrak{S}_M(N)$  and  $\mathfrak{S}(N)$  differ only in the local factors at the primes dividing  $M$ . For  $\Delta = -4$ ,  $M = 4$ , one has  $\frac{M}{\varphi(M)^2} \mathfrak{S}_M(N) = \frac{1}{2} \mathfrak{S}(N)$ , recovering Proposition 3.3.  $\square$

#### DATA AVAILABILITY AND ACKNOWLEDGEMENTS

The verification script `certify.py`, the exploratory scan and their outputs are archived at doi:10.5281/zenodo.23137626 and available at

<https://github.com/Ruqing1963/bilateral-two-squares-sieve>.

During the preparation of this work, the author utilized the Claude language model and the `mpmath` Python library for computer-assisted interval arithmetic verification of the sieve integrals in Section 4. All analytical derivations and proofs were independently audited and verified by the author.

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